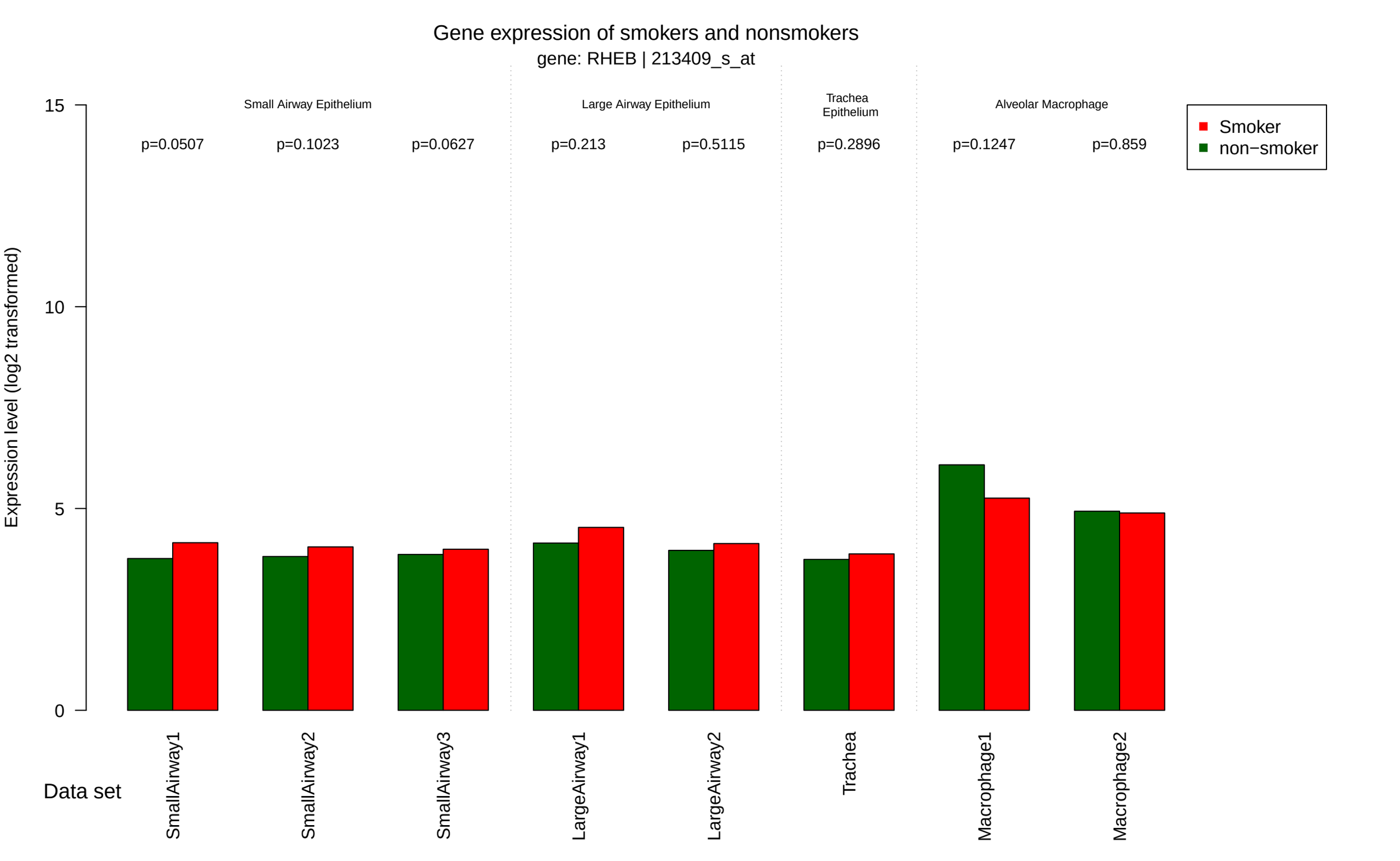
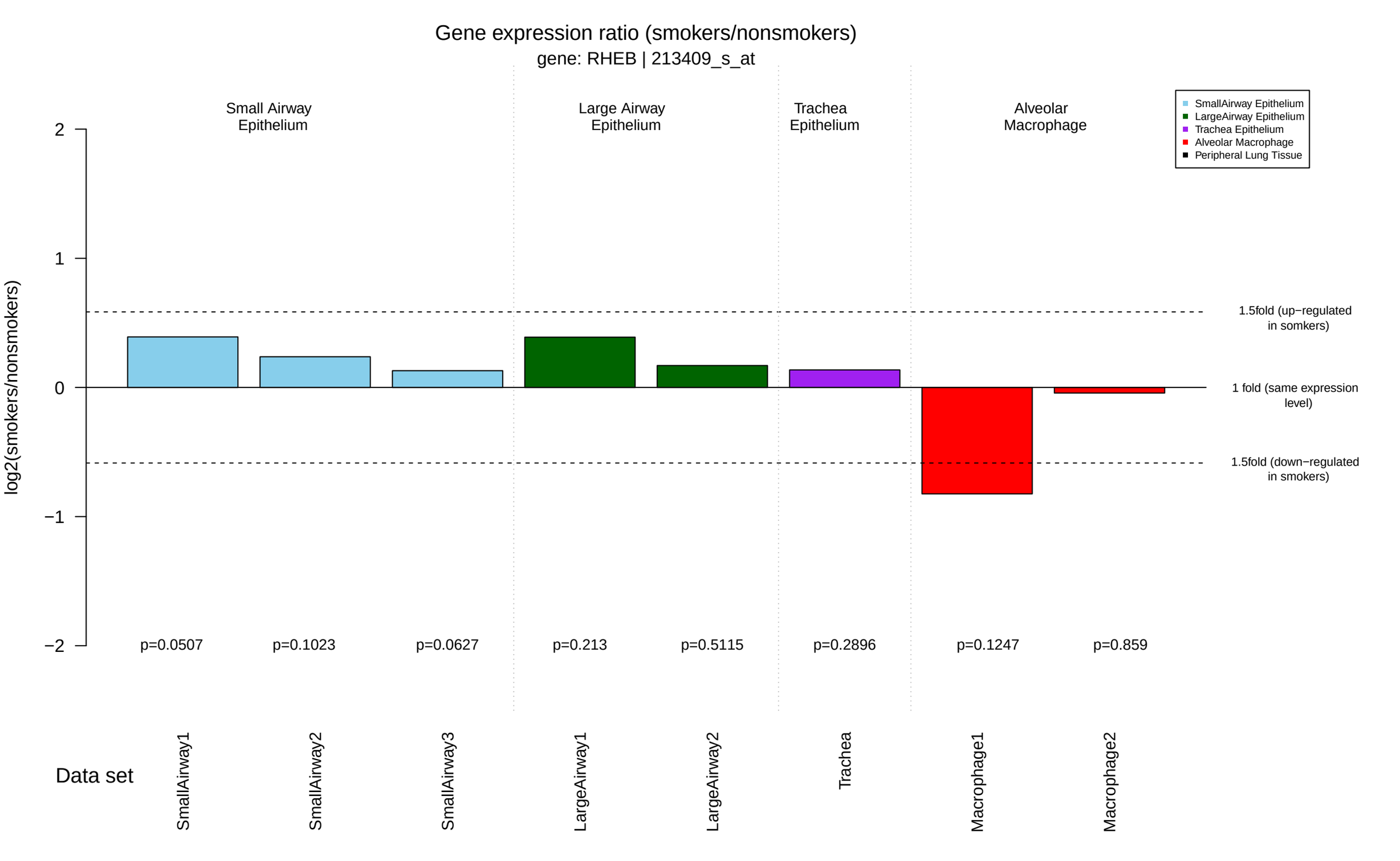
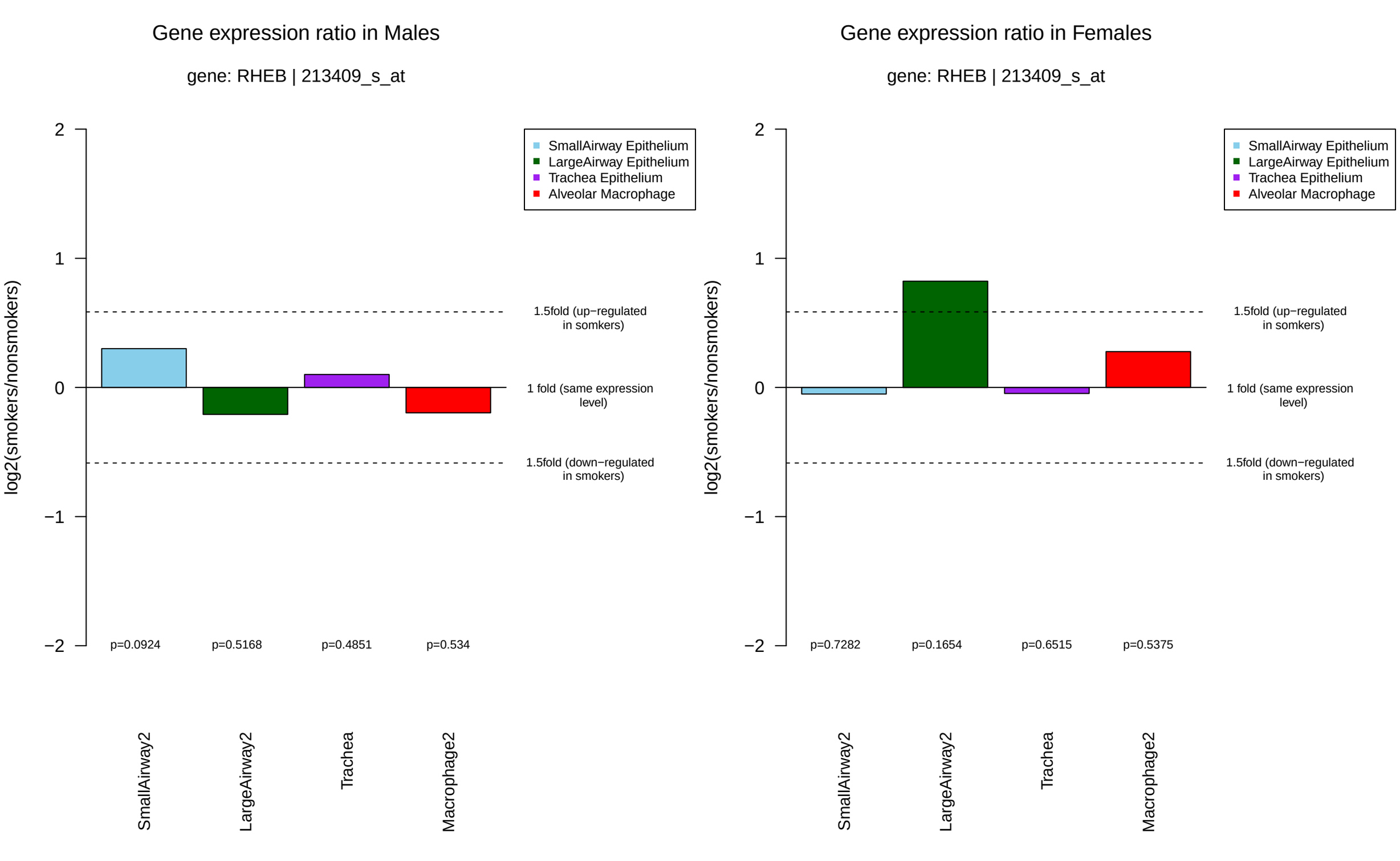
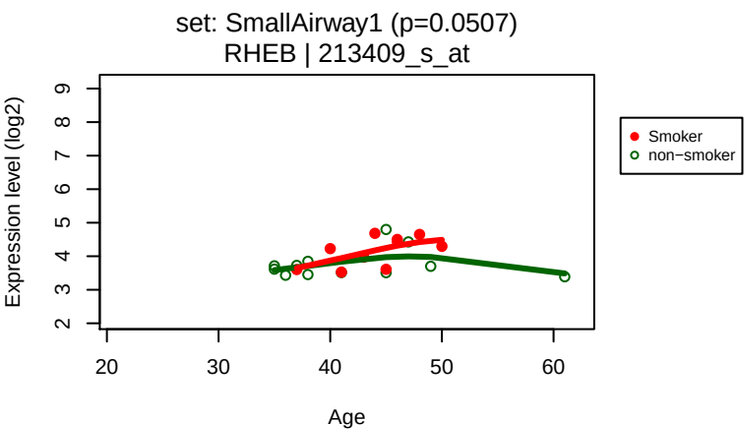
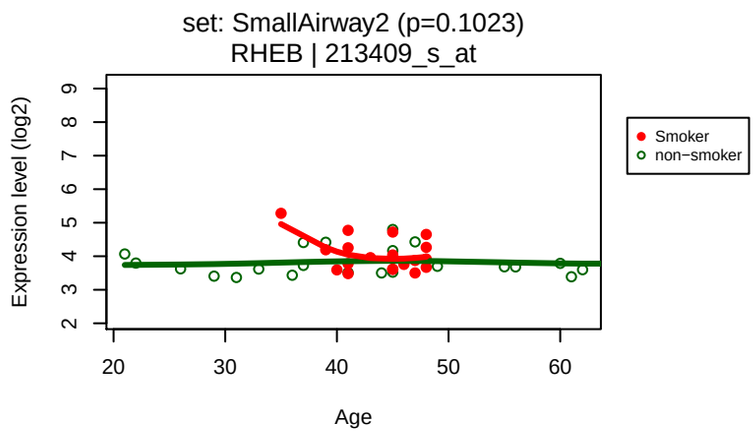
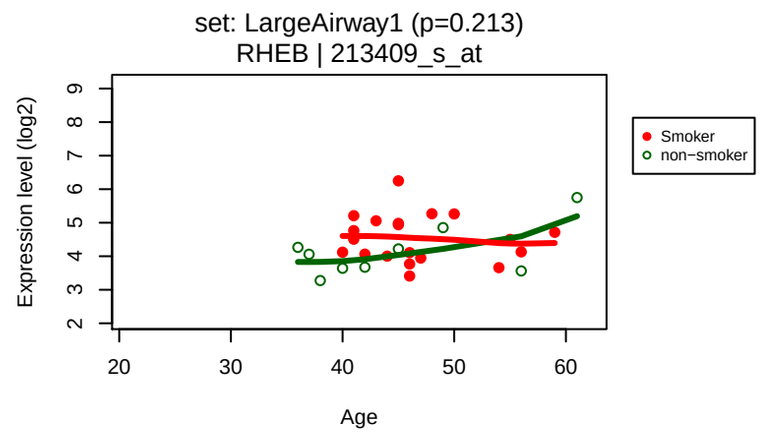

Gene expression of smokers and nonsmokers
gene: RHEB | 213409_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0507
p=0.1023
p=0.0627
p=0.213
p=0.5115
p=0.2896
p=0.1247
p=0.859
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

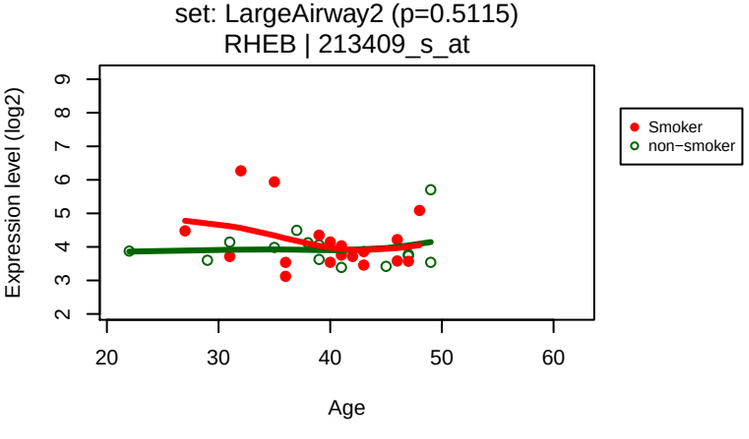

Gene expression ratio (smokers/nonsmokers)
gene: RHEB | 213409_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0507
p=0.1023
p=0.0627
p=0.213
p=0.5115
p=0.2896
p=0.1247
p=0.859
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

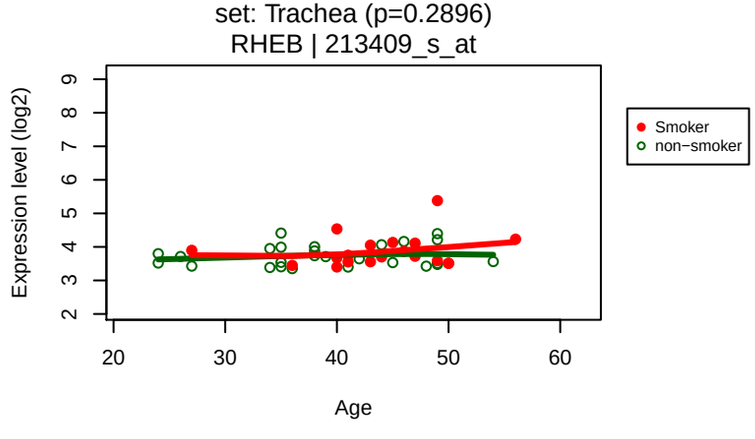

Gene expression ratio in Males
gene: RHEB | 213409_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0924
p=0.5168
p=0.4851
p=0.534
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RHEB | 213409_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7282
p=0.1654
p=0.6515
p=0.5375
SmallAirway2
LargeAirway2
Trachea
Macrophage2

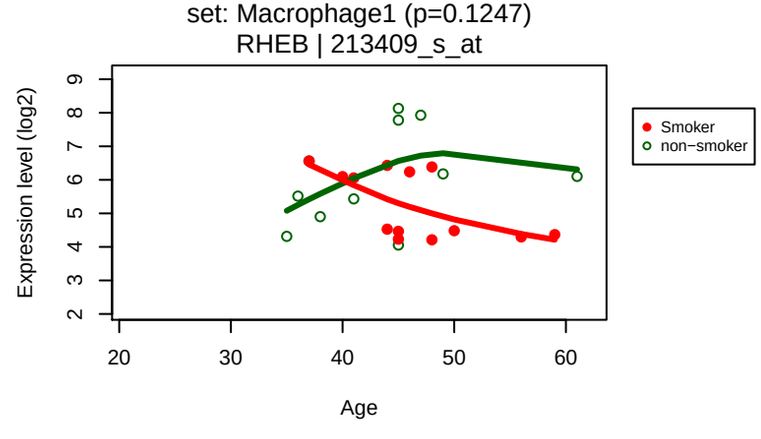

set: SmallAirway1 (p=0.0507)
RHEB | 213409_s_at
Expression level (log2)
Age
Smoker
non−smoker

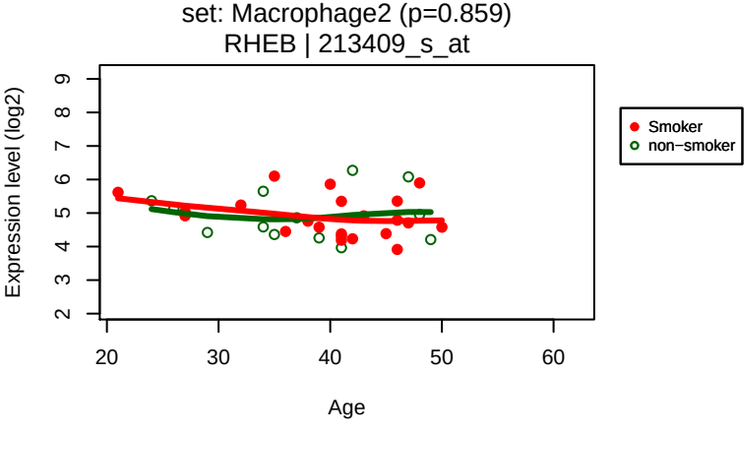

set: SmallAirway2 (p=0.1023)
RHEB | 213409_s_at
Expression level (log2)
Age
Smoker
non−smoker

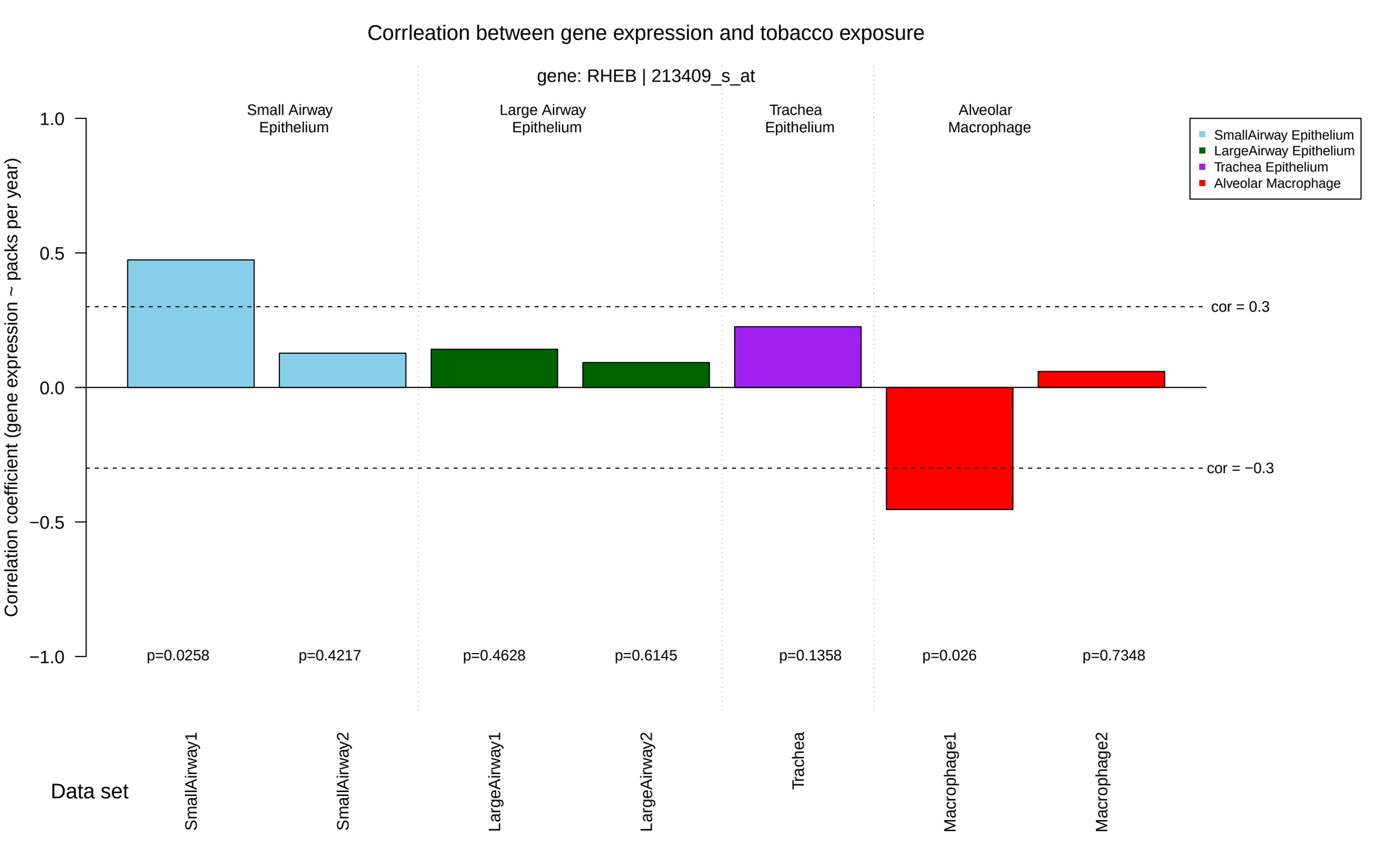

set: LargeAirway1 (p=0.213)
RHEB | 213409_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5115)
RHEB | 213409_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2896)
RHEB | 213409_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1247)
RHEB | 213409_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.859)
RHEB | 213409_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RHEB | 213409_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0258
p=0.4217
p=0.4628
p=0.6145
p=0.1358
p=0.026
p=0.7348
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2